

Balance of payments
and
Exchange rate



Larale
Mushtag

Balance of payments

The balance of payments is a record of the transactions that one country has with the rest of the world. It shows all the various payments and receipts arising from a country's involvement in international trade. It consists of the following three components:

- * The current account
- * The financial account
- * The capital account

* Balancing item, also known as net errors of omissions

The **financial account** is the part of the balance of payments that records the movement of funds into and out of a country, such as direct investment in the form of building a factory or portfolio investment in the form of the buying or selling of government bonds.

The **capital account** is the part of the balance of payments that records capital movements, in terms of various assets and liabilities, into and out of a particular country, such as money brought into and taken out of a country by migrants.

In addition to the three accounts, the balance of payments also includes a

balancing item. The balance of payments should eventually balance when all of the various accounts are included. However, it is possible that statistical discrepancies may prevent this from happening and so a balancing item is used to ensure that the accounts, when added together, equals zero.

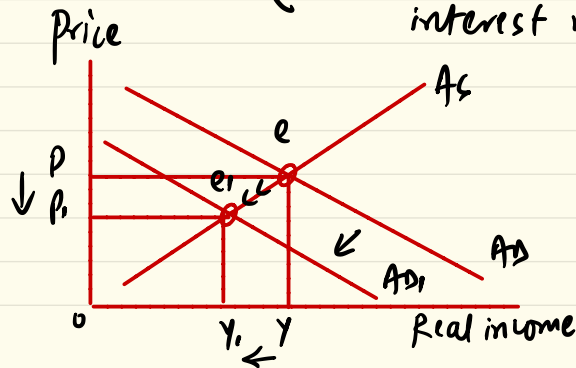
Expenditure Switching and reducing (Dampening) policy



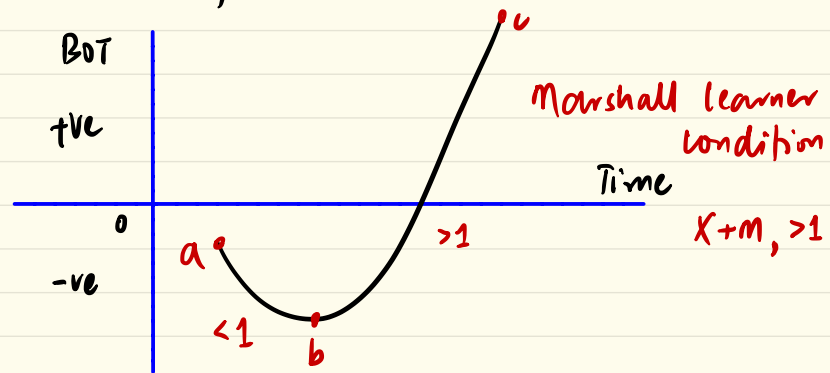
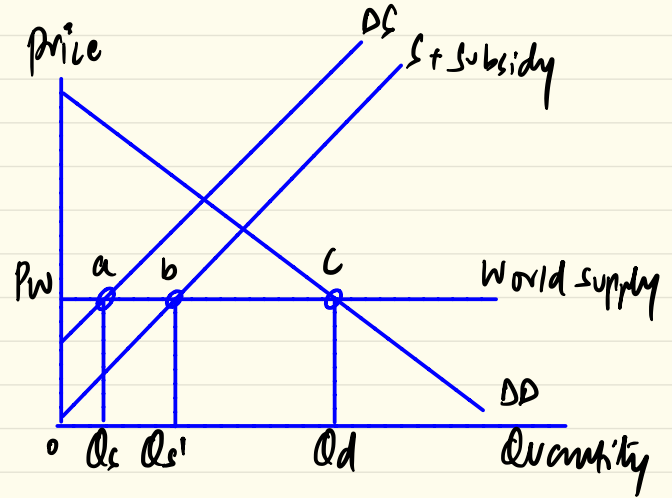
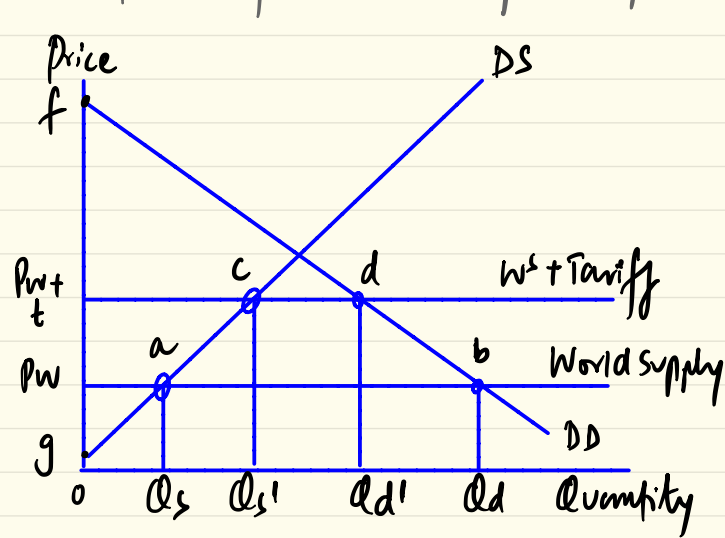
- * Tariff
- * Subsidy
- * Devaluation / Depreciation



- * Fiscal (Tax $\uparrow$ and Government spending $\downarrow$)
- * Monetary (Money supply $\downarrow$ interest rate $\uparrow$)



Diagrams for Switching policy



Expenditure reducing policies are those which are designed to reduce the demand for all products in an economy, that is, they are intended to reduce the demand for imported goods, but the effect is that the demand for all goods, including domestically produced goods, will be reduced.

Deflationary fiscal policy, to reduce expenditure, can include an increase in taxation and/or a reduction in government expenditure. Deflationary monetary policy, to reduce expenditure, can include an increase in interest rates and/or a reduction in the money supply.

Expenditure switching policies are designed to switch demand away from some products and towards others, for example, they could intend to decrease the demand for imports and increase the demand for exports.

There are number of protectionist methods that can be used to decrease the demand for imports and increase the demand for exports, including tariff, quotas, subsidies, exchange controls, embargoes and administrative restrictions.

Exchange rate

There are a number of ways of measuring the price of a currency. It can, for example, be measured in terms of how much it costs in another currency. It can also be measured in terms of what can it buy in another country or countries and in terms of a basket of currencies.

Nominal and real exchange rates

The nominal foreign exchange rate is the price of one currency in terms of another currency. For example, 160 Pakistani rupees may buy one US\$.

Economists also measure real exchange rates. A real exchange rate takes price changes as well as exchange rate changes into account to assess changes in the competitiveness of a country's products in global markets. A fall in the country's foreign exchange rate would be expected to make its exports more price competitive. However, if the country is experiencing a relatively high inflation rate, export prices may actually be increasing. A real exchange rate compares the relative price of two countries' products:

$$\text{Real exchange rate} = \frac{\text{nominal exchange rate} \times \text{Domestic price index}}{\text{foreign price index}}$$

The real exchange rate may rise as a result of appreciation or an increase in the country's relative inflation rate. The price of products in the country will increase relative to those in another country or countries. More imports could be purchased for a given quantity of exports. As the country's products are more expensive relative to its competitors, the country is likely to import more and export less. The real exchange rate has more of an impact on the current account balance than the nominal exchange rate as it reflects international price competitiveness.

Trade weighted exchange rate

In practice, a country trades with the number of other countries and it is possible that its currency may rise in value against some countries while falling against others. To gain an impression of the general change in a country's foreign exchange rate, economists measure trade weighted exchange rate.

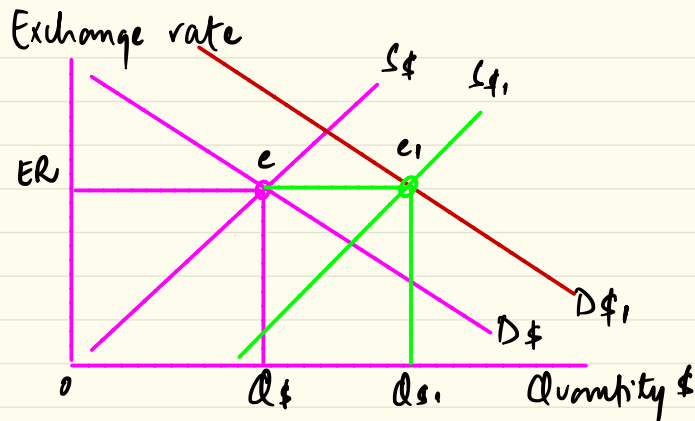
A trade weighted exchange rate is a measure, in index form, of the price of a currency against a basket of currencies. These are weighted according to the relative importance of the countries in the country's trade. For example, if India undertakes three times as much trade with China as it does with the US, the Chinese renminbi will be given three times as much weight in the calculation as the US \$.

Fixed Exchange rate

Governments generally aim to avoid large fluctuations in the external value of the currency. This will be easier to achieve with a fixed exchange rate system. In this type of exchange rate system, the external value of a currency is determined by the government of the country, that is, the government decides to fix the exchange rate at a particular level, rather than leaving the rate to be decided by market forces as in a floating exchange rate system.

The advantage of a fixed exchange rate system is that a government can intervene in the foreign exchange market on a regular basis in order to maintain a particular rate of exchange with other currencies. If there is a floating exchange rate, this will be more difficult to achieve because the rate of exchange with other currencies will be determined by changes in the demand for, and the supply of, the currency on foreign exchange markets.

However, a disadvantage of a fixed exchange rate system is that a government will need to ensure that it has sufficient reserves in order to be able to intervene in the foreign exchange market to maintain a particular exchange rate.

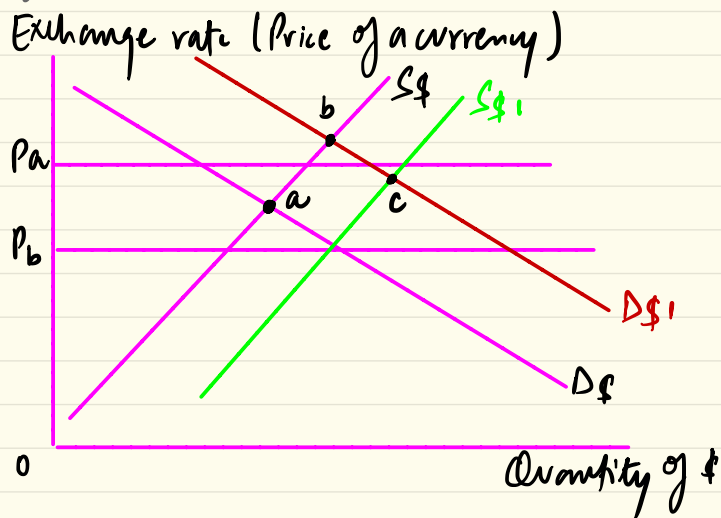


Managed exchange rate systems

A managed exchange rate system is a combination of floating and fixed exchange rates. A government may decide to allow the exchange rate of a currency to be determined by market forces, but only to an extent; it will aim to reduce any large fluctuations in a floating exchange rate system through a degree of intervention and this is why it is known as a managed float.

The government will decide the minimum and maximum rates between which the value will range. If market forces determine the rate above or below those maximum or minimum values, the government will intervene to buy or sell the currency to ensure

that its value stays within the predetermined limits.



Devaluation and Revaluation: If an exchange rate is lowered in a fixed exchange rate system, it is called a devaluation. If the exchange rate is increased in a fixed exchange rate system, it is called a revaluation.

Changes in Exchange rate

* The demand for a country's exports from other countries

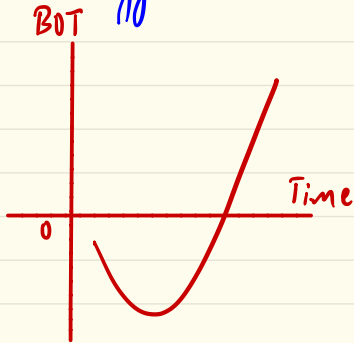
- * The demand for imports into a country from other countries
- * Relative inflation rates in different countries, affecting the international competitiveness of goods and services that are traded between different countries.
- * The quality and reliability of products that are traded internationally
- * Relative interest rates in different countries, which can be a major factor in the movement of 'hot money' from one country to another
- * Changes in the cost of production in different countries
- * Changes in the levels of productivity in different countries
- * Differences in the state of technology in different countries

Marshall lerner analysis

The Marshall lerner condition states for a reduction in the external value of a currency to be successful, the sum of the price elasticity of demand for exports and the price elasticity of demand for imports needs to be greater than one. The Marshall lerner condition, in relation to the price elasticity of demand for a country's exports and imports, is useful in terms of judging whether a reduction in the value of a country's exchange rate is likely to be successful in improving a country's balance of payment situation.

Juene analysis: A country's exchange rate could be lowered to encourage an increase in exports and a decrease in imports. This aim, however, might not

happen immediately and it is possible that there may be a period of time when the situation becomes worse before it gets better. This situation is known as the J-curve effect.



Initially the current account situation worsens as a result of the reduction in the external value of the currency, and the situation only improves after a certain period of time has elapsed.

Q1 Devaluation of a country's currency will reduce a persistent balance of payments deficit on its current account in goods and services in the short run but this will inevitably lead to high level of inflation in the long run.

Evaluate this statement.

Ans:- * Define devaluation (under fixed / managed floating exchange rate)

* A persistent balance of payments deficit on the country's current account in goods and services will create problems because it has to be financed. If it is not financed by positive currency flows on the financial account due to direct investment or portfolio investment, it will have to be financed by reducing reserves of foreign currency. This particular problem will only apply to a country that has adopted a fixed exchange rate.

* In this situation the difference between the short run and the long run will be determined by the time taken to reduce the deficit and will also be measured by the final impact on the general price level, that is the rate of inflation.

* Devaluation, $\times$ prices $\downarrow$, $\uparrow$ prices $\uparrow$, this can create BOT surplus, assuming, more exports will be sold and fewer goods will be imported. This would help to reduce BOT deficit.

* To some extent, the ability of domestic suppliers to reduce a country's current account deficit in the short run depends on how domestic suppliers respond to the opportunity to sell more exports at lower price. We can measure this response by calculating price elasticity of supply of domestic producers.

* A devaluation might not solve a balance of payments deficit on a country's current account in goods and services in the short run because domestic supply of goods might be inelastic, this means domestic producers cannot increase output sufficiently to meet the additional demand for exports.

* Devaluation, $\uparrow$ prices, $\uparrow$ costs, raw material, cost push inflation, diagram.
Further analysis of a devaluation on the general price level could be provided. For example, analysis might consider the proportion of imports accounted for by raw materials and how this might affect the cost of goods and services provided in an economy and how an increase in costs might lead to cost push inflation.

Evaluation:-

* In the long run, it is necessary to consider both the price elasticity of demand for imports and price elasticity of demand for exports. The Marshall-Lerner condition states that the sum of the two elasticities must be greater than one to achieve a successful outcome.

* While the longrun impact on the current account is dependent upon respective export/import price elasticities of demand, it is also important to recognize that if raw material such as oil constitute a significant proportion of the value of imports then the price elasticity of demand for imports is likely to be low.

* The longrun impact on inflation will be influenced by the current state of the economy when the devaluation takes place. For example, if there is significant amount of spare capacity (deflationary gap), the effect on aggregate demand of a lower price of exports might be limited.

X prices ↓, M prices ↑, PED/PES elastic, $X(TE) > M(TE)$, BOT surplus, AD ↑, spare capacity | deflationary gap exist, AS ↑, mild inflation!

* It could also be noted that a government which chooses to use a policy of devaluation is perhaps prioritising the need to correct a balance of payment disequilibrium at the possible expense of other key macroeconomic goals such as controlling inflation.

* Based on the evidence/analysis provided, it is not clear whether a devaluation of the currency will solve a persistent current account deficit in either the shortrun or longrun. Similarly, the impact on the general price level depends on a number of variables which are difficult to predict or measure accurately.

Devaluation

X prices ↓
X Qty ↑

M prices ↑
M Qty ↓

X(TE)?

M(TE)?

Inelastic

Elastic

Inelastic

Elastic

X(TE) ↓

X(TE) ↑

M(TE) ↑

M(TE) ↓

↓
Low PES

↓
High PES

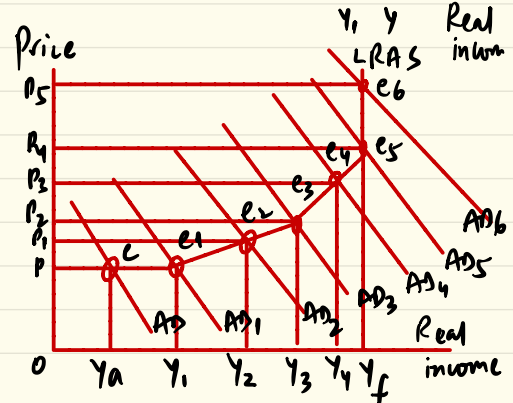
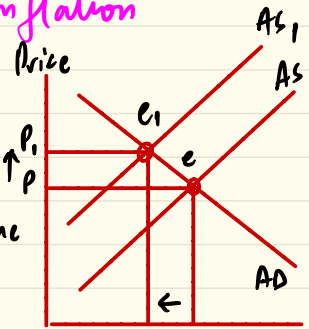
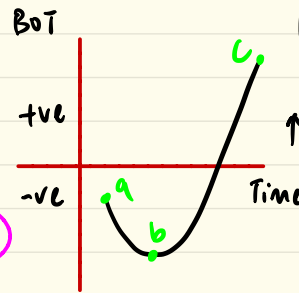
↓
Low PED

↓
High PED

↓
cost push inflation

Shortrun, BOT:- deficit
cost push inflation

Longrun:- BOT:- surplus
Demand pull inflation



Q2 Revaluation of a country's currency will decrease a persistent balance of payments surplus on its current account in goods and services in the shortrun but this will inevitably lead to low level of inflation in the longrun.

Evaluate this statement.

Ans:- * Define Revaluation (under fixed / Managed floating exchange rate)

$X_{prices} \uparrow$

$M_{prices} \downarrow$

$X_{Qty} \downarrow$

$M_{Qty} \uparrow$

$X(TE) ?$

$M(TE) ?$

* Shortrun, PED/PES, inelastic, BOT:- Surplus
Aggregate supply $\uparrow$, cost push inflation $\downarrow$

* Longrun, PED/PES, elastic, BOT:- deficit
Aggregate demand $\downarrow$
Controls demand pull inflation

$X(TE) \uparrow$
inelastic

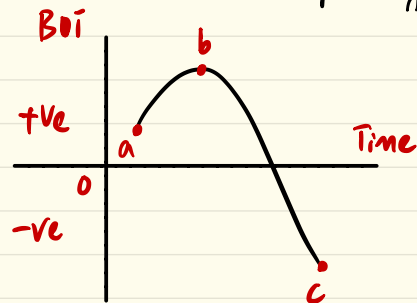
$X(TE) \downarrow$
elastic

$M(TE) \downarrow$
inelastic
*

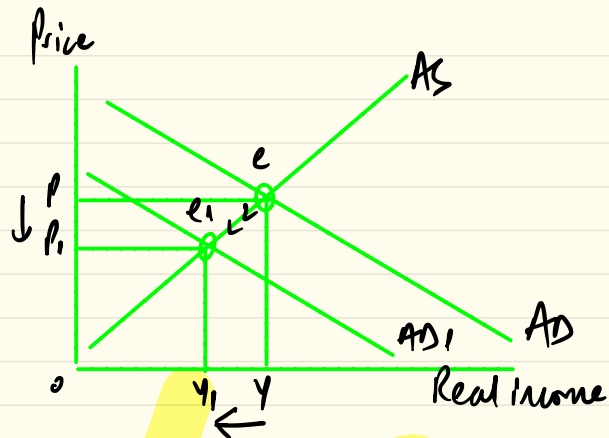
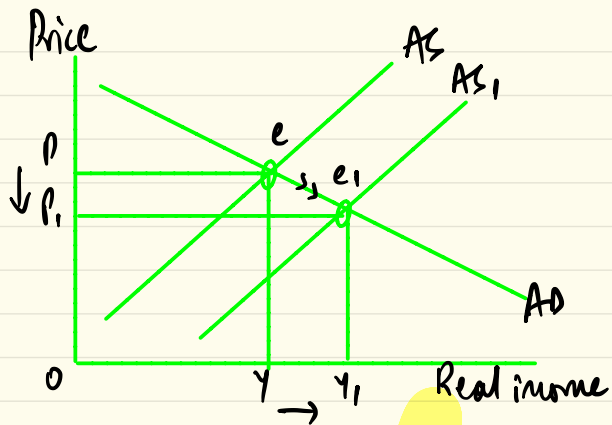
$M(TE) \uparrow$
elastic

BOT surplus

BOT deficit



* $M(TE) \downarrow$, reduces cost push inflation



- * Reverse Marshall learner condition, combined elasticity of demand for exports and imports needs to be inelastic, then only BOT will be in surplus
- * Revaluation will be beneficial when the economy is closer to full employment, then trade deficit will reduce aggregate demand and controls inflation.
- * Revaluation will be beneficial, considering the economy has comparative advantage
- * Revaluation controls cost push with elasticity being inelastic and control demand pull with elasticity being elastic.

Q3 Expenditure Switching policy will be successful in correcting current account deficit but will also cause high rate of inflation? 20

Q4 Expenditure dampening policy will be successful in correcting current account deficit but will also cause significant unemployment? 20